

Related Work and Positioning for an Angular Manifold Routing Preprint

This note is meant to be inserted, adapted, or mined for language in a future preprint. Its purpose is to acknowledge the nearest adjacent literature while keeping the novelty claim narrow and defensible. The central empirical claim of the current project is not that geometry has never been used in routing, but that **normalized angular partitioning can induce emergent routing sparsity**, with a measured sublinear occupancy law and hardware-locality consequences.

1. How to position the contribution

A safe positioning statement is:

While prior work has explored geometric representation learning, hyperspherical similarity spaces, and mixture-of-experts routing, we are not aware of prior work that combines normalized hyperspherical embeddings, deterministic angular sector routing, emergent routing sparsity via bucket compression, and an empirical routing occupancy scaling law with hardware-locality implications.

2. What existing work is closest

Hyperbolic and non-Euclidean representation learning

A large literature studies hyperbolic and other curved representation spaces for hierarchical data, graphs, taxonomies, and language structure. The 2024 perspective paper *Brain-Inspired AI with Hyperbolic Geometry* argues that hyperbolic geometry may provide more efficient representations for brain-like and AI systems, but it is a representation-geometry argument rather than a routing-law architecture. It does not present fixed angular bucket routing or an occupancy scaling law.

Standard mixture-of-experts routing

Mainstream MoE systems route tokens with a learned gate that scores experts and selects a sparse subset. This is the dominant framing in the field: experts plus router plus load-balancing. The overlap with the present work is sparse compute. The difference is that routing in MoE is normally *learned*, while the current project emphasizes sparsity that *emerges from geometric partitioning*.

Hyperspherical MoE-adjacent work

The closest older neighbor I found is *On the Representation Collapse of Sparse Mixture of Experts* (2022). That paper computes routing scores between tokens and experts on a low-dimensional hypersphere using dimensionality reduction and L2 normalization. This is a meaningful overlap. However, it still uses a learned routing score and does not propose deterministic angular sector buckets or an empirical law of the form $E(K) \approx c * K^\alpha$ with $\alpha < 1$.

Recent spherical routing variants

A recent example is *SLER-IR: Spherical Layer-wise Expert Routing for All-in-One Image Restoration* (2026), which uses spherical expert routing ideas in image restoration. It is close in spirit, but still relies on learned activation behavior and does not focus on routing occupancy scaling or hardware-locality consequences.

Geometry-aware aggregation

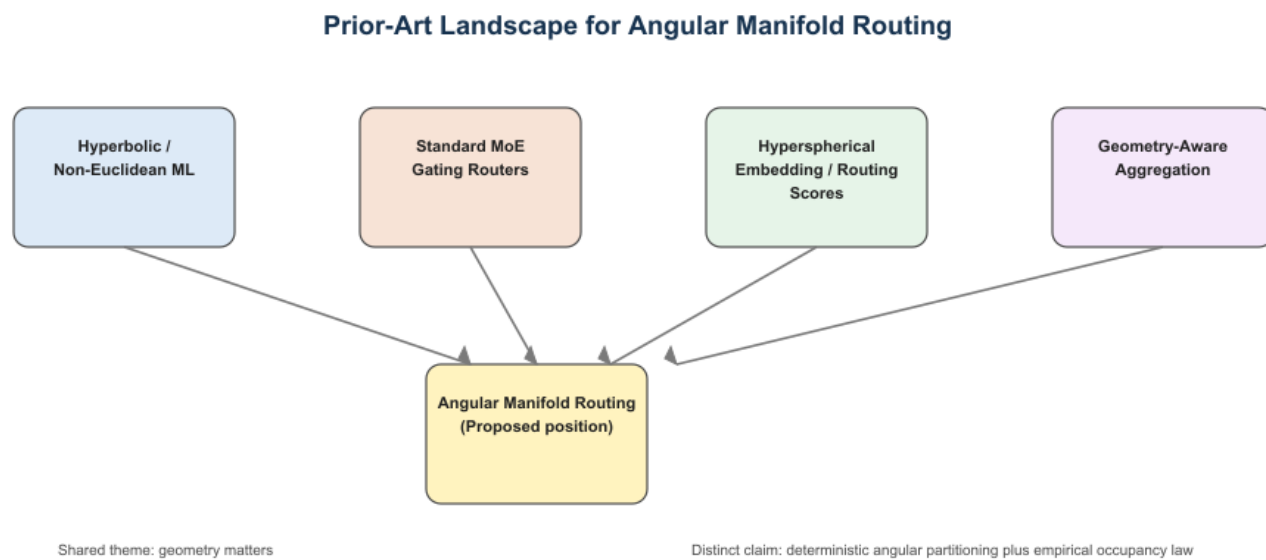
Geometry-Preserving Aggregation for Mixture-of-Experts Embedding Models (2026) studies angular and radial structure when combining expert outputs. It is useful prior art for manifold-aware expert systems, but it concerns aggregation rather than the routing mechanism itself.

Latent routing-space methods

Work such as *L2R: Low-Rank and Lipschitz-Controlled Routing for Mixture-of-Experts* (2026) improves routing by learning a better latent routing space. The overlap is the idea that routing geometry matters. The difference is that routing remains a learned transformation rather than a deterministic angular partition of the embedding manifold.

3. Visual prior-art landscape

The diagram below is not a claim of exhaustive coverage. It is a communication aid showing how the current project sits relative to nearby clusters of work.



4. Summary comparison table

Area of prior work	Main idea	Closest overlap	Key difference from this project
Hyperbolic / non-Euclidean ML	Curved spaces represent hierarchy efficiently	Geometry affects efficiency and structure	Focus is on representation geometry, not deterministic angular routing
Standard MoE routing	Learned gate selects experts	Sparse conditional compute	Router is trained; sparsity is not claimed to emerge from fixed angular partitioning
Hyperspherical MoE methods	Normalized embeddings with cosine-style routing scores	Sphere geometry enters routing computation	Geometry is used as a scoring space, not a fixed sector router with an occupancy law
Spherical routing variants	Domain-specific spherical routing ideas	Explicit spherical routing language	Still learned / similarity-driven; no measured bucket-occupancy scaling law
Geometry-aware aggregation	Preserve angular and radial structure when combining expert outputs	Angular/radial manifold awareness	Aggregation geometry rather than routing geometry
Latent routing-space methods	Learn improved routing geometry in latent space	Routing quality depends on geometric structure	Routing remains a learned function rather than deterministic partitioning

5. Suggested references to include

Chi, Z. et al. (2022). *On the Representation Collapse of Sparse Mixture of Experts*. NeurIPS. This is the closest older neighbor because it places routing scores on a low-dimensional hypersphere.

Joseph, A., Francis, N., and Balay, M. (2024/2025 version). *Brain-Inspired AI with Hyperbolic Geometry*. Use this as broad context for the importance of non-Euclidean geometry in AI and brain-inspired modeling.

Peng, S. et al. (2026). *SLER-IR: Spherical Layer-wise Expert Routing for All-in-One Image Restoration*. Useful to cite as a recent spherical-routing-adjacent paper in another domain.

Kachuee, S. and Sharifkhani, M. (2026). *Geometry-Preserving Aggregation for Mixture-of-Experts Embedding Models*. Useful to cite when discussing manifold-aware expert systems that are not themselves routing laws.

Yang, M. et al. (2026). *L2R: Low-Rank and Lipschitz-Controlled Routing for Mixture-of-Experts*. Good contrast paper for learned latent routing geometry.

A standard MoE reference such as *Switch Transformers* should also be included to anchor the baseline learned-gating paradigm.

6. Final wording suggestion

If you want one paragraph to place near the end of the Related Work section, use something like this:

Prior work has shown that geometry can matter both for representation learning and for expert routing. In contrast to learned routing functions and hyperspherical scoring approaches, the present work studies a

deterministic angular partitioning mechanism on normalized embeddings. The main empirical claim is that routing sparsity can emerge from bucket compression itself, producing a sublinear occupancy law and downstream hardware-locality benefits.